

Tutorial 2 (Iterative and Number Theoretic Algorithms)

1. Explain iterative GCD algorithm and analyze its time and space complexity.
2. Explain iterative algorithm for Fibonacci terms and analyze its time and space complexity.
3. Explain bubble sort algorithm and analyze its time and space complexity.
4. Explain selection sort algorithm and analyze its time and space complexity.
5. Explain insertion sort algorithm and analyze its time and space complexity.
6. Explain Extended Euclidean algorithm and analyze its time and space complexity.
7. Find out the GCD of 55 and 34 using Extended Euclidean algorithm. Also find the values of coefficients a and b such that $55a + 34b = \gcd(55, 34)$
8. Find out the GCD of 161 and 28 using Extended Euclidean algorithm. Also find the values of coefficients a and b such that $161a + 28b = \gcd(161, 28)$
9. Trace the following data using bubble sort, selection sort and insertion sort algorithms.
 - i. $A = \{6, 5, 4, 3, 2, 1\}$
 - ii. $A = \{24, 63, -5, 8, 1\}$
 - iii. $A = \{8, 5, 7, 3, 2\}$
 - iv. $A = \{5, 2, 4, 6, 1, 3\}$
10. Solve the following modular equations using Chinese remainder theorem.

$x \equiv 2 \pmod{3}$	$x \equiv 1 \pmod{3}$
a) $x \equiv 3 \pmod{5}$	b) $x \equiv 3 \pmod{5}$
$x \equiv 2 \pmod{7}$	$x \equiv 6 \pmod{7}$
$x \equiv 0 \pmod{2}$	$x \equiv 1 \pmod{2}$
c) $x \equiv 2 \pmod{3}$	d) $x \equiv 3 \pmod{5}$
$x \equiv 3 \pmod{5}$	$x \equiv 6 \pmod{7}$
$x \equiv 1 \pmod{3}$	
$x \equiv 1 \pmod{4}$	
e) $x \equiv 1 \pmod{5}$	
$x \equiv 0 \pmod{7}$	
11. Find all integers x that leave remainders 1, 2, 3 when divided by 9, 8 and 7 respectively.
12. Check whether the following numbers are probably prime or composite using Miller-Rabin Primality test algorithm.

a) 561	d) 2465
b) 1105	e) 1009
c) 1729	f) 2017
13. State Fermat's Theorem. Check whether the following numbers are probably prime or composite using Fermat's Theorem.

a) 7	c) 143
b) 53	d) 9
14. Evaluate the following using the concept of modular exponentiation.

a) $3^{100} \pmod{29}$	c) $1314^{343} \pmod{159197}$
b) $2^{35} \pmod{561}$	d) $33677^{12007} \pmod{159197}$
15. Find the modular inverse of the following using Fermat's Theorem.

a) 5 mod 7	c) 6 mod 17
b) 5 mod 11	d) 35 mod 3